

4(a)	- acts as 'return' (conductor) for signal - shielding from noise/crosstalk/interference <i>Two sensible suggestions, 1 mark each.</i>	B2
4(b)	- small bandwidth (there is) noise/interference/crosstalk - large attenuation/energy loss - reflections due to poor impedance matching <i>Two sensible suggestions, 1 mark each.</i>	B2
4(c)	attenuation = $190 \times 14 \times 10^{-3}$ (= 2.66 dB)	C1
	ratio/dB = $(-10 \lg(P_2/P_1))$	C1
	$2.66 = -10 \lg(P_{\text{OUT}}/P_{\text{IN}})$ $P_{\text{OUT}}/P_{\text{IN}} = 0.54$	C1
	fractional loss = $1 - (P_{\text{OUT}}/P_{\text{IN}}) = 1 - 0.54$ = 0.46	A1
	or	
	$2.66 = 10 \lg(P_{\text{IN}}/P_{\text{OUT}})$ $P_{\text{IN}}/P_{\text{OUT}} = 1.85$	(C1)
	fractional loss = $(P_{\text{IN}} - P_{\text{OUT}})/P_{\text{IN}} = (1.85 - 1)/1.85$ = 0.46	(A1)